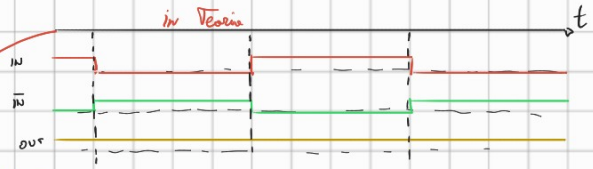


$OUT = IN + \overline{IN} = 1$





eventi termici

se il cambiamento è quello in giallo è il sistema più semplificato

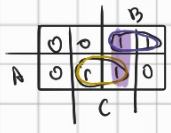
tempo di reazione da una porta logica che può variare, non è uguale → come il tubo dell'acqua

Alea = errore derivante l'uscita dovuta alla propagazione dei segnali

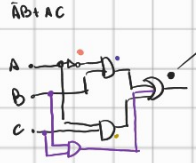
↓

la soluzione è il CLOCK, il quale aspetta che il sistema arrivi a regime

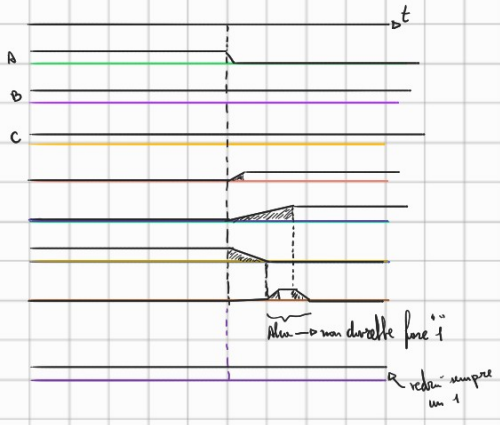
si può vedere anche nelle mappe di Karnaugh

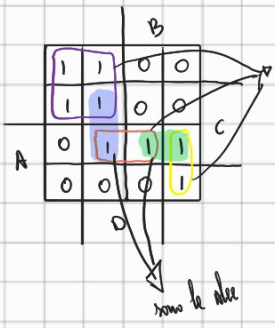


$\overline{A}B + AC$

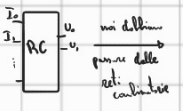


↓ devo fare il diagramma temporale

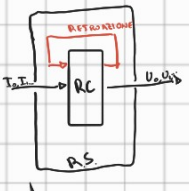




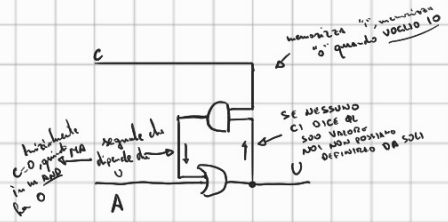
soluzione minima



a una rete sequenziale

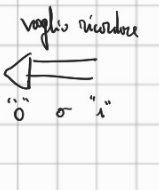


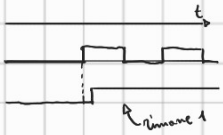
Esempio:



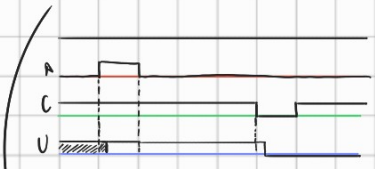


↳ diagramma temporale semplificato: sono una FOTOGRAFIA del mondo





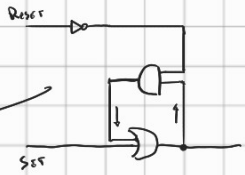
→ elemento di memoria che memorizza un "1"



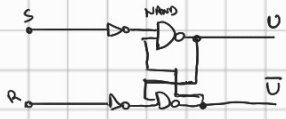
non mi piace avere queste variabili A e C

CAMBIO

Usando De Morgan vediamo un cambiamento

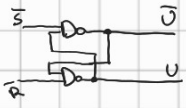


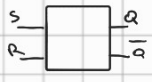




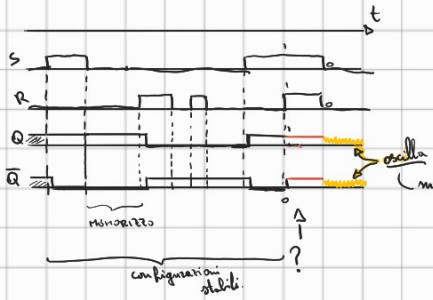


A	B	NAND
0	0	1
0	1	1
1	0	1
1	1	0









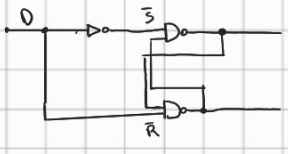
succede contemporaneamente solo se il circuito è PERFETTO, nella vita reale non esiste la perfezione.

S	R	Q	$\bar{Q}$
0	0	Q	$\bar{Q}$
0	1	0	1
1	0	1	0
1	1	NON AMMESSO	

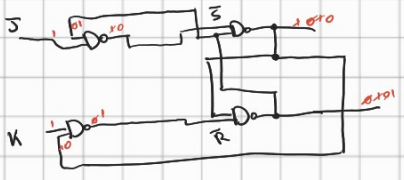
continuo a mantenere al suo interno i valori che sono stati memorizzati

TABELLA DI ECCITAZIONE

Un altro FLIP FLOP



D	$\bar{S}$	$\bar{R}$
0		
1		



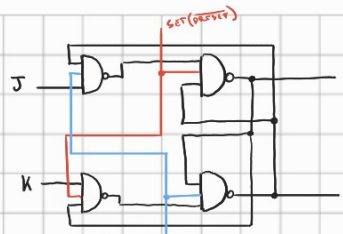
J	K	Q	$\bar{Q}$
0	0	Q	$\bar{Q}$
0	1	0	1
1	0	1	0
1	1	OSCILLA!!!	

← comando di memorizzazione

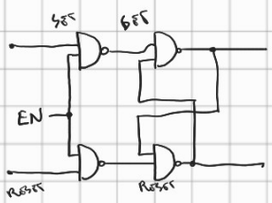
← forza la memorizzazione di "01" o "10"

circuito simmetrico

← comando di reset "01"



$\overline{\text{RESET (CLEAR)}}$



EN	S	R	Q	$\bar{Q}$
0	–	–	Q	$\bar{Q}$
1				

EN	D	Q	$\bar{Q}$
0	–	Q	$\bar{Q}$
1			

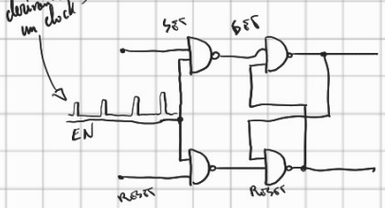
P.F.D FLOP

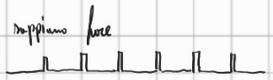
EN	J	K	Q	$\bar{Q}$
0	–	–	Q	–
1				

F.F. J K

derivante dal sistema interno

derivante da un clock





ck = CLOCK

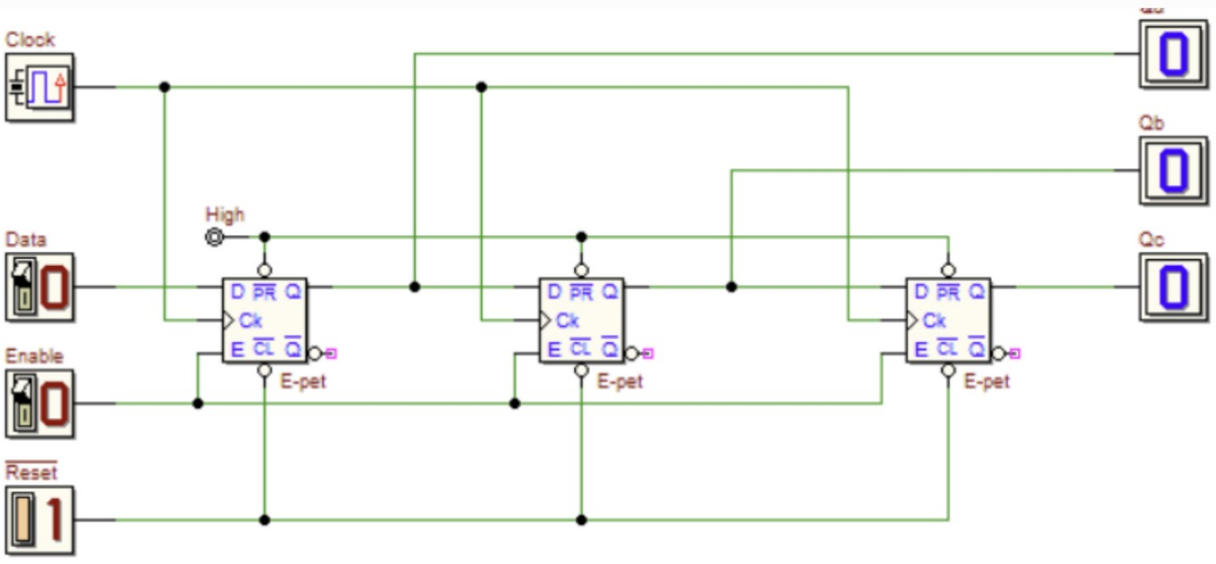
F.F J K NEGATIVE EDGE TRIGGERED

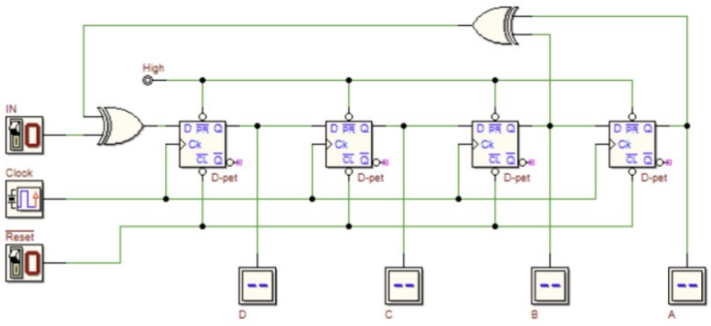
1→0

PET POSITIVE EDGE TRIGGERED

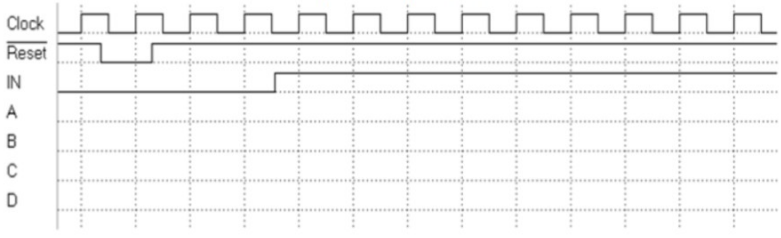
0→1

guardare le SLIDE del prof. + LIBRO

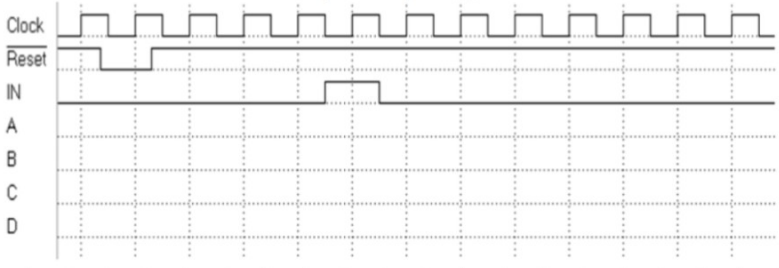




Sequenza temporale (A):



Sequenza temporale (B):



PR = 1

